



Technical Service Bulletin

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Dedicated Fuel Pump Lubricating Oil Filter Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK60 CM2350 K116
- QSK60 CM2350 K121
- QSK78 CM2350 K126

Description of Change



Figure 1, New Dedicated Fuel Pump Lube Oil Filter Introduction.

A new dedicated lubricating oil filter for the fuel pump is now available.

New fuel pump lubricating oil filter provides direct filtration for the fuel pump lubricating oil supply immediately before the inlet feed drilling of the fuel pump. This is accomplished through:

- A New fuel pump drive adapter
- Fuel pump drive adapter integrated filter head
- Fleetguard® 20 micron lubricating oil filter, Part Number LF17560.
- New engine control module (ECM) calibration code(s) with auto-detect new engine main wiring harness (left bank)
- New extension wiring harness

Additional features include:

- Additional pre-lube engine cranking circuit switch
- Fuel pump lubricating oil filter outlet pressure sensor and new fault code(s)
 - See Figure 2 below.

Cummins Description	Error	Fault Code	SPN	FMI	Lamp Color	Notes
Engine Fuel Pump Oil Pressure Sensor Circuit - Voltage Above Normal or Shorted to High Source	FUEL_PUMP_OIL_PRESS_OOR_HIGH_ERROR	5119	520754	3	Amber	
Engine Fuel Pump Oil Pressure Sensor Circuit - Voltage Below Normal or Shorted to Low Source	FUEL_PUMP_OIL_PRESS_OOR_LOW_ERROR	5121	520754	4	Amber	
Engine Fuel Pump Oil Pressure - Data Valid But Below Normal Operating Range - Most Severe Level	FUEL_PUMP_OIL_PRESS_RPM_DRT_ERROR / FUEL_PUMP_OIL_PRESS_TRQ_DRT_ERROR	7313	520754	1	Red	Will be used for speed or torque ^a derate
Engine Fuel Pump Oil Pressure - Data Valid But Below Normal Operating Range - Moderately Severe Level	Fuel pump oil press warning error	7314	520754	18	Amber	This will be used for warning only. No derate.

17r00270

Figure 2, New Fault Codes.

Reason for Change

Improve modular common rail fuel system (MCRS) fuel pump life.

Customer Communication

Cummins Inc. recommends that Master Rebuild Centers retrofit the dedicated lubricating oil filter. Factory option(s) on applicable engines during engine rebuild.

Service Instructions

Install new fuel pump lubricating oil filter extension wiring harness.

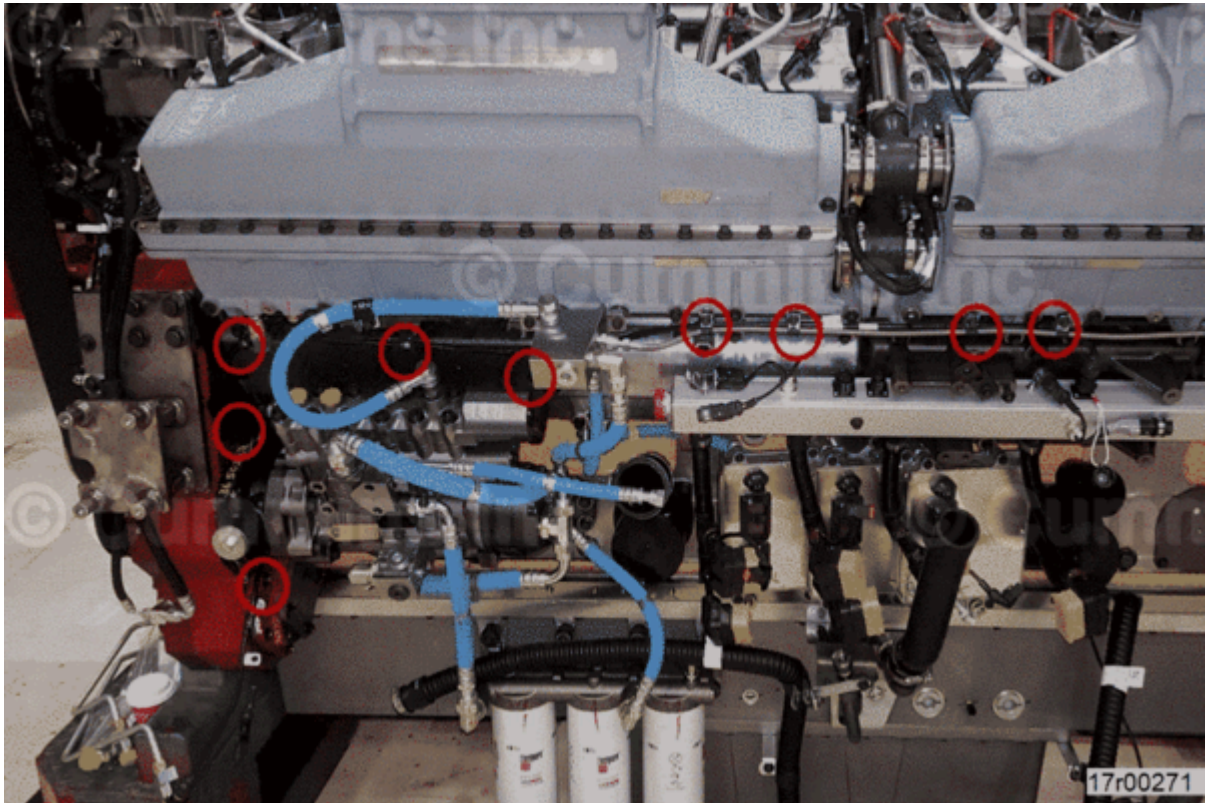


Figure 3, Extension Wiring Harness P-clip Locations.



Figure 4, Extension Wiring Harness Connection.

1. Connect new wiring harness extension to main left bank wiring harness. See Figure 4 above.

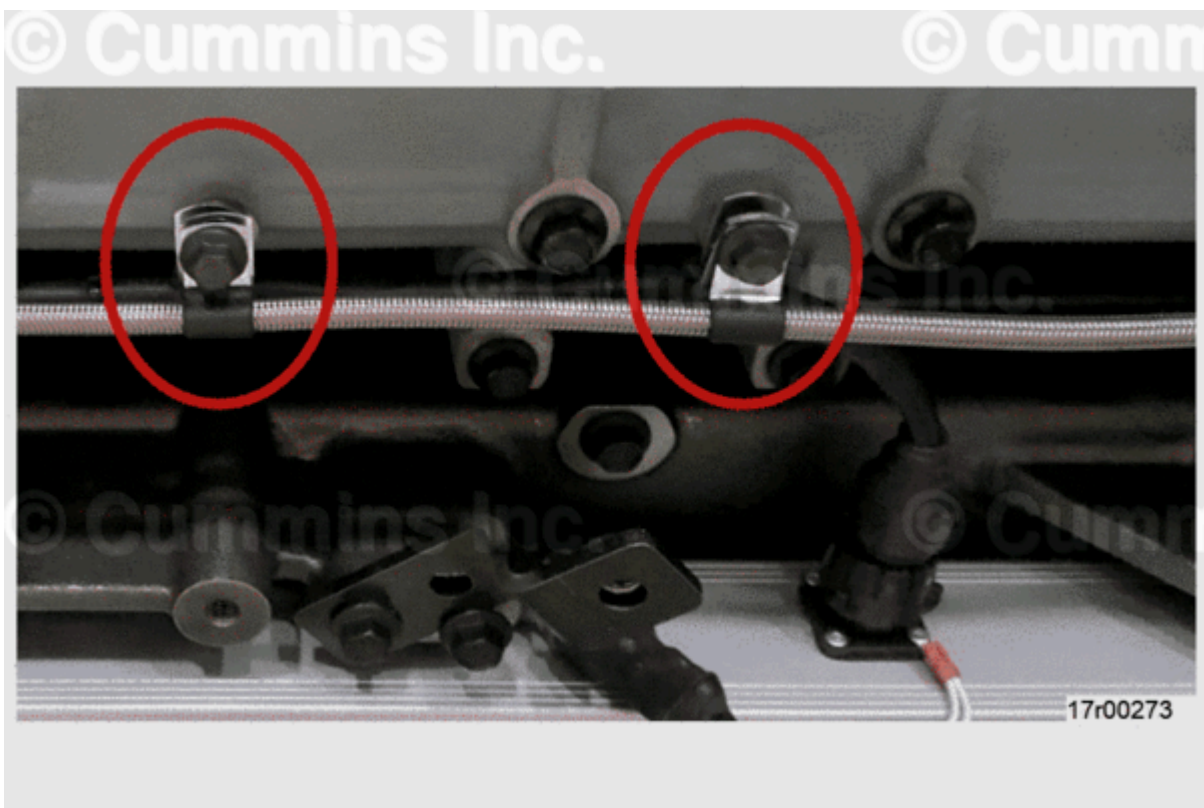


Figure 5, P Clip Locations.

2. Install two new p-clips with capscrews to aftercooler bolt holes in an inboard orientation. Verify wiring harness is **not** interfering with surrounding components. See Figure 5 above.
3. Torque capscrews to 33N•m [24ft-lb]

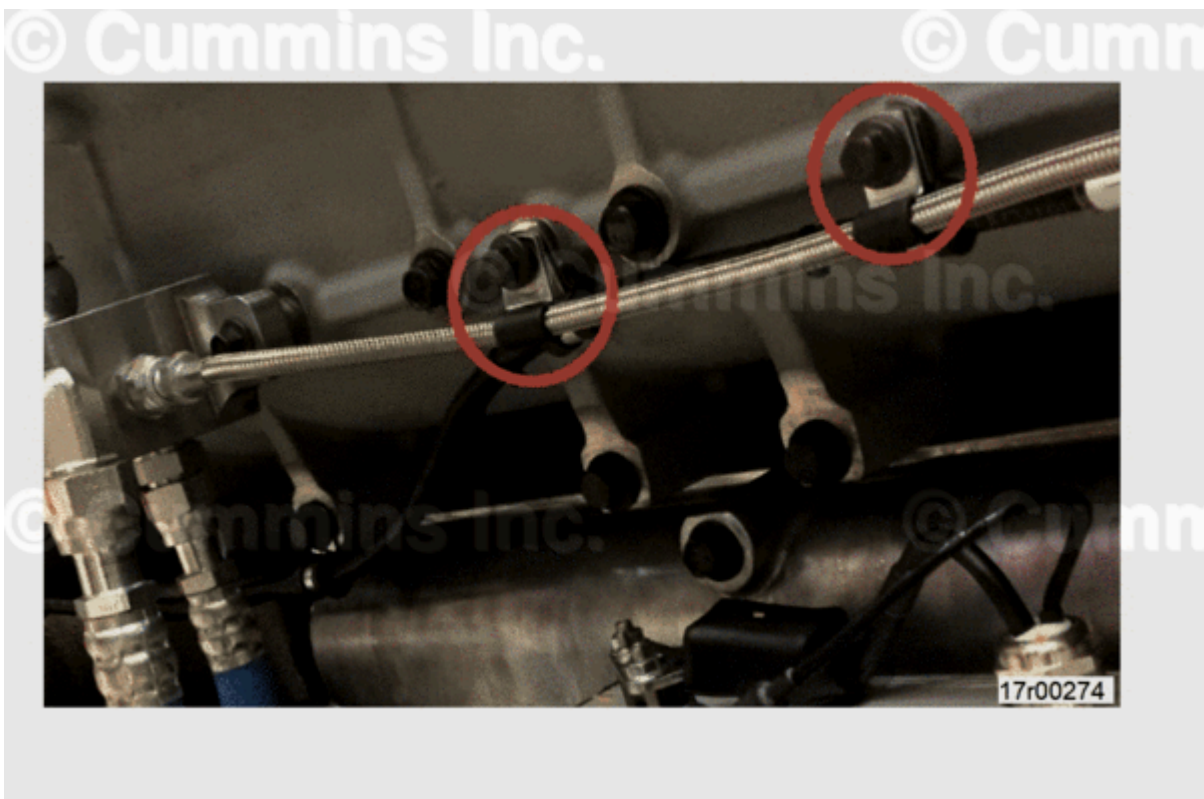


Figure 6, P Clip Locations.

4. Install two new p-clips with capscrews to aftercooler bolt holes in an inboard orientation. Verify wiring harness is **not** interfering with surrounding components. See Figure 6 above.
5. Torque capscrews to 33N•m [24ft-lb]



Figure 7, P Clip Locations.

6. Install new p-clip into bottom of aftercooler bolt hole in an inboard orientation. Verify wiring harness is **not** interfering with surrounding components. See Figure 7 above.
7. Torque capscrews to 46N•m [33ft-lb]



Figure 8, P Clip Locations.

8. Install new p-clip into bottom of aftercooler bolt hole in an inboard orientation. Verify wiring harness is **not** interfering with surrounding components. See Figure 8 above.

9. Torque capscrews to 46N•m [33ft-lb]



Figure 9, P Clip Locations.

10. Install new p-clip into bottom of aftercooler bolt hole in an inboard orientation. Verify wiring harness is **not** interfering with surrounding components. See Figure 9 above.

11. Torque capscrews to 46N•m [33ft-lb]

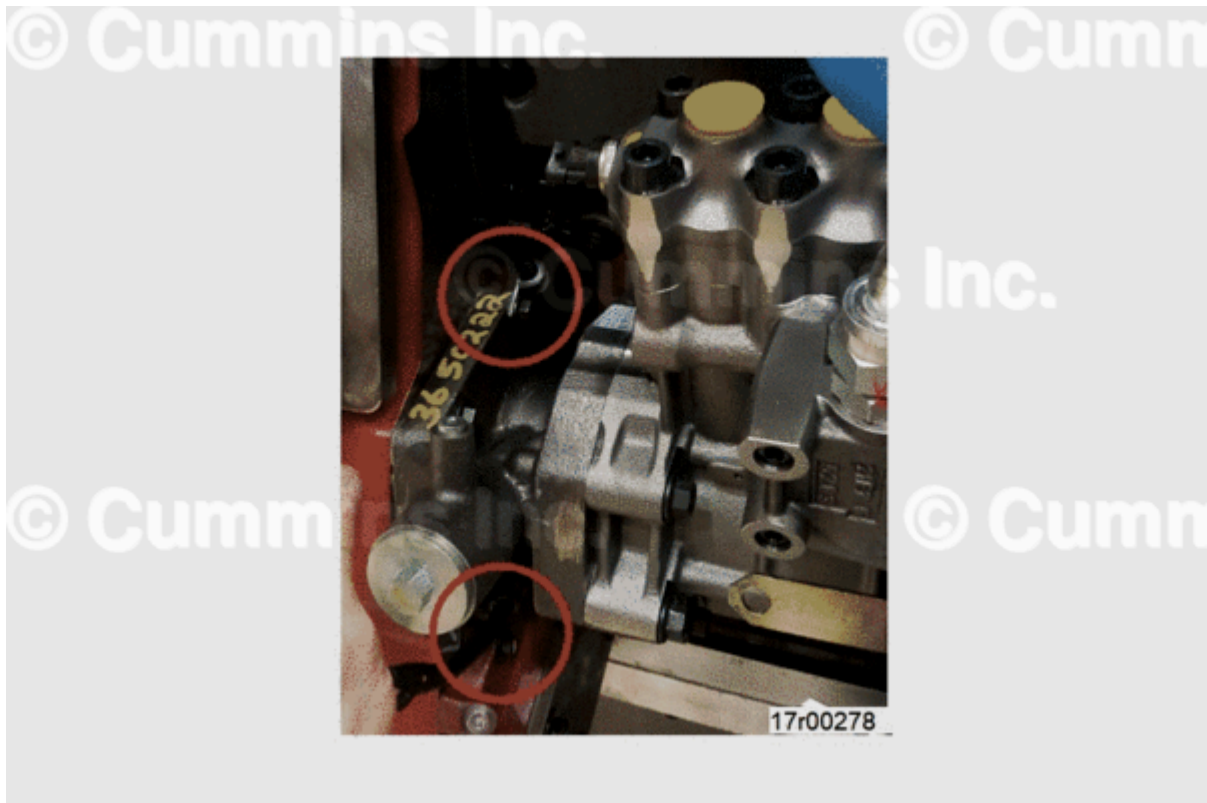


Figure 10, P Clip Locations.

12. Install two new p-clips with capscrews to aftercooler bolt holes in an inboard orientation. Verify wiring harness is **not** interfering with surrounding components. See Figure 10 above.

13. Torque capscrews to 45N•m [33ft-lb]

Installation of Fuel Pump Lubricating Oil Filter Head.

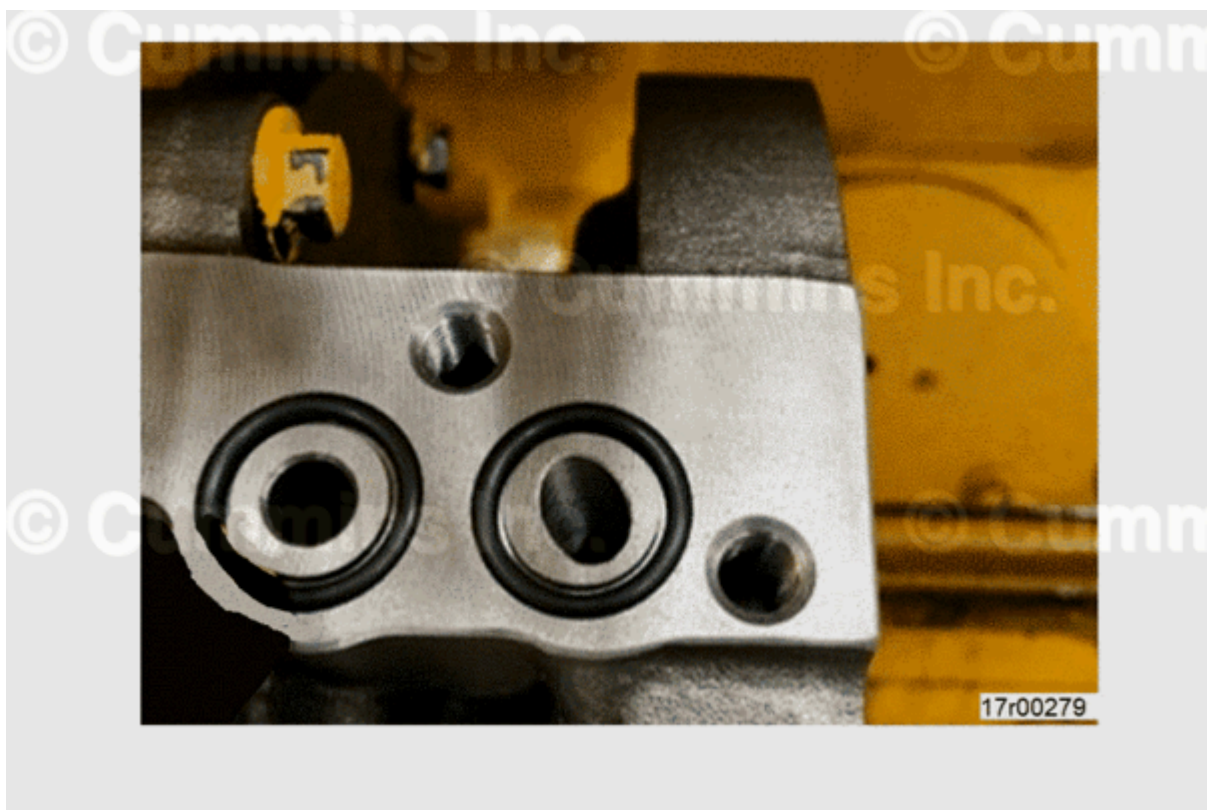


Figure 11, O-ring Ports.

1. Install o-rings in accessory drive support. See Figure 11 above.

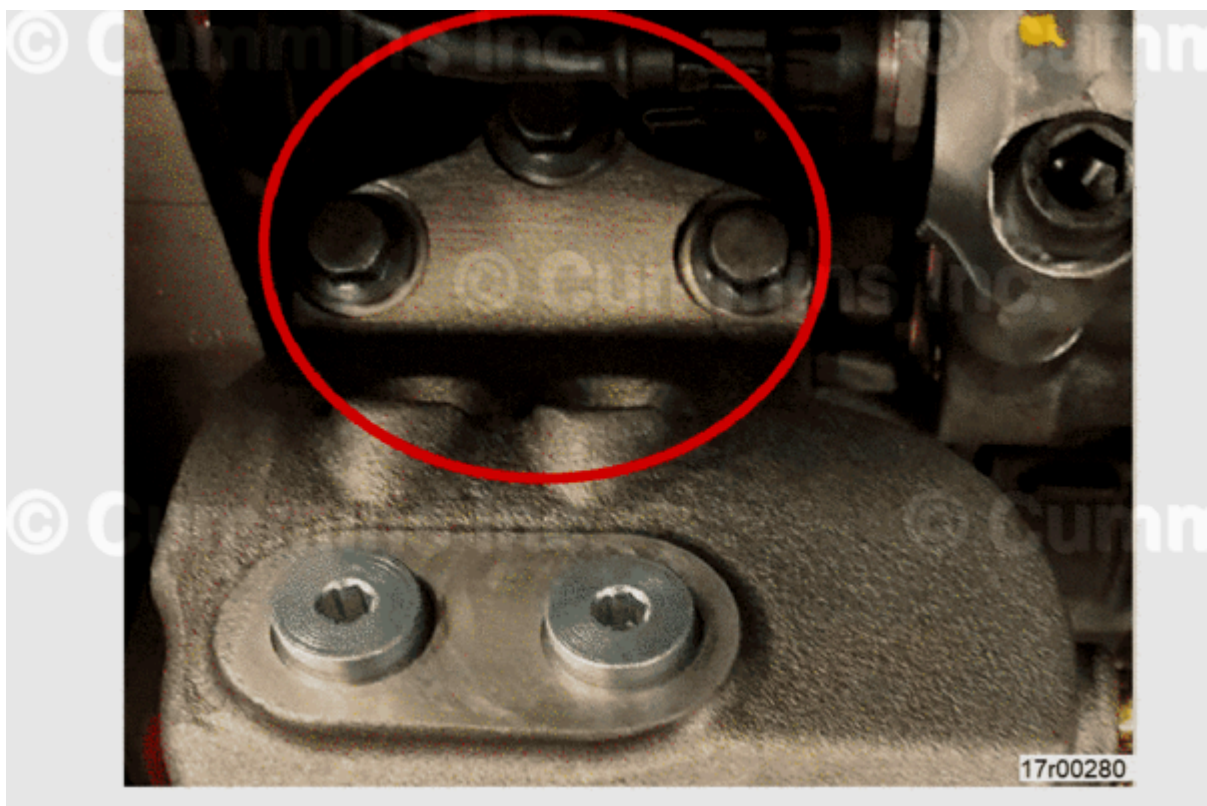


Figure 12, Capscrew Locations for Installation of Fuel Pump Lubricating Oil Filter Head.

2. Install fuel pump lubricating oil filter head to fuel pump with three capscrews. See Figure 12 above.

3. Torque capscrews to 46N•m [33ft-lb]

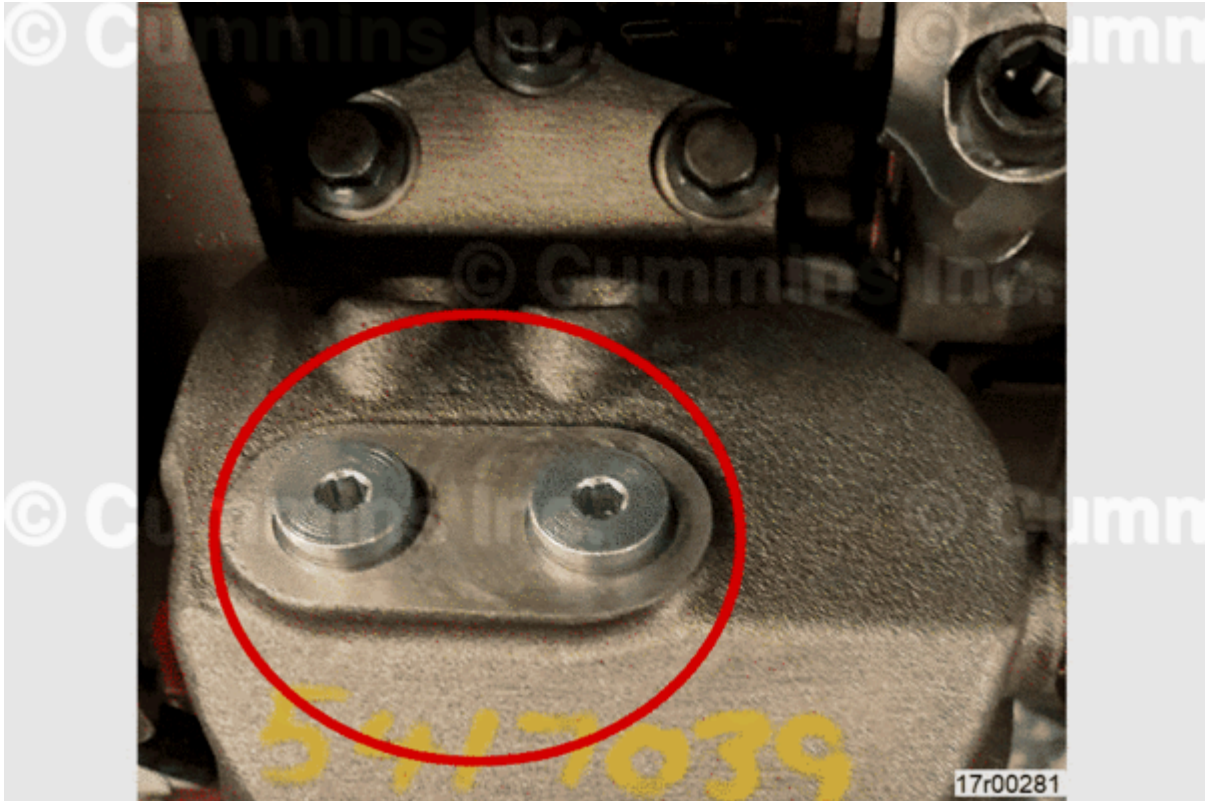


Figure 13, Port Plug Locations on Fuel Pump Lubricating Oil Filter Head..

4. Install port plugs into top of fuel pump lubricating oil filter head. See Figure 13 above.

5. Torque to 11N•m [8ft-lb]

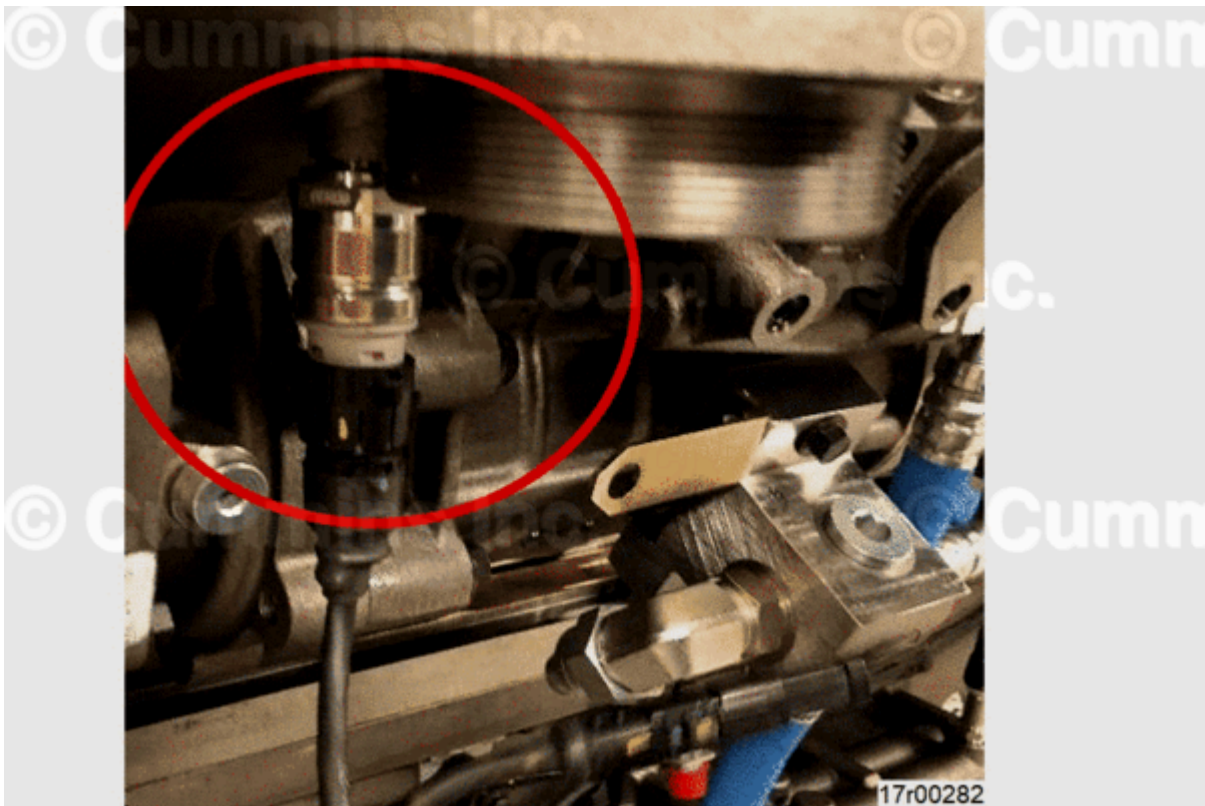


Figure 14, Primary Lubricating Oil Pressure Sensor Location.

6. Install primary lubricating oil pressure sensor behind filter location in lubricating oil fuel pump lubricating oil filter head. See Figure 14 above.

7. Torque to 11N•m [8ft-lb]

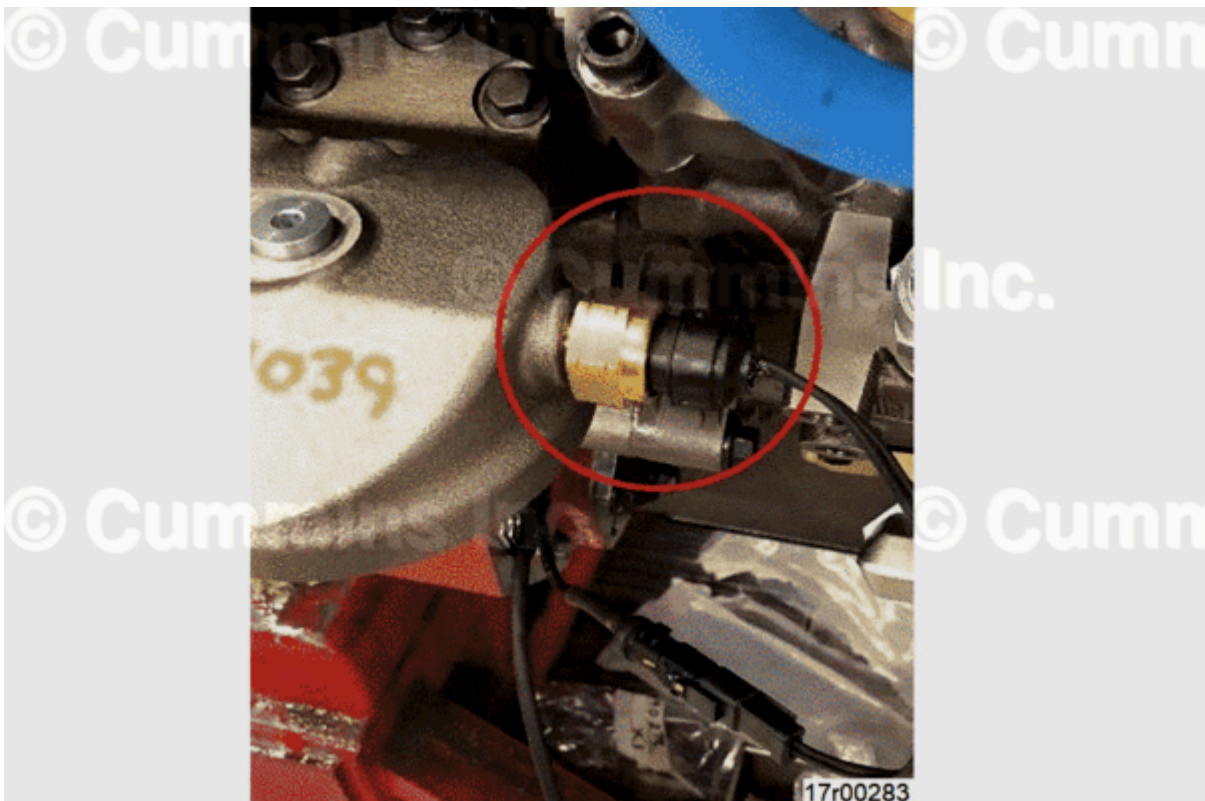


Figure 15, Secondary Lubricating Oil Pressure Sensor Location.

8. Install secondary lubricating oil pressure sensor behind filter location in location lubricating oil filter head. See Figure 15 above.

9. Torque to 11N•m [8ft-lb]



Figure 16, Fuel Pump Lubricating Oil Filter Canister.

10. Install fuel pump lubricating oil filter hand tight. See Figure 16 above.



Figure 17, Lubricating Oil Pressure Wiring Harness Sensor Connection.

11. Connect extension wiring harness to pressure sensors. See Figure 17 above.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts			
Option Number	Part Description	Part Number	Quantity
FO6701	FILTER,FUEL PUMP OIL	-	-
	DECAL,INFORMATION	4323855	1
	FILTER,LUBRICATING OIL	4323856 (LF17560)	1
	HARNESS,ETR CNT MDL WRG	5374882	1
	SCREW,HEX FLANGE HEAD C	3093923	4
	SENSOR,PRESSURE	3408560	1
	SCREW,HEX FLANGE HEAD C	3902460	1
	SWITCH,PRESSURE	4924925	1
	HEAD,LUB OIL FILTER	5417039	1
	SEAL,O RING	3328463	2
	PLUG,THREADED	3678921	2
	SCREW,HEX FLANGE HEAD C	3916069	3
DP6724	DRIVE,FUEL PUMP	-	-
	SUPPORT,ACCESSORY DRIVE	5416815	1
	SCREW,HEX FLANGE HEAD C	3089016	6
	GASKET,ACC DRIVE SUPPOR	3332297	1
	HARNESS,ETR CNT MDL WRG	5374880	1
	HARNESS,ETR CNT MDL WRG	5374881	1

Part Compatibility

- Only released at this time for QSK60 T2 Industrial applications with space claim for on engine mounted configuration. A QSK60 remote mount option as well as options for QSK50 and QSK38 are under development.
- Factory release does **not** include Komatsu PC4000/PC8000/WA1200, NHL haul truck, Hitachi EX3600/EH5000, or Cat 6090.

Production Statu

Implemented for production. See Table 2.

Table 2, Production Information

ESN First	Build Date ¹	Plant
33213266	7 March 2018	Daventry Engine Plant

¹Engine build date can be found on the engine dataplate.

Document History

Date	Details
2019-1-3	Module Created

Last Modified: 09-May-2019
